



The Psychological Risks of BCI-LLM Cognitive “Enhancement”

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Abstract Allen Coin & Veljko Dubljević discuss the potential implications of a brain-computer interface (BCI) that sends input signals to a calculator which would then “...transmit the result of the calculation to the subject in such a way that the subject could immediately mentally note, verbalize, or record the result” [1]. In other words, the system sends an output, presumably via deep brain stimulation (DBS), which induces thought-like experiences. They characterize this system as being able to enhance cognition and restore the mathematical capacity and self-esteem of those with dyscalculia. However, many reports suggest that DBS and BCI technology can have negative unintended psychological consequences. I introduce a related conceptual model of a “BCI-LLM system” to better illustrate the possible negative

unintended psychological consequences of *contentful* DBS, or neural stimulations that produce meaningful sensations, regardless of whether true cognitive enhancement occurs. It is possible, for example, that a subject can be *fooled* into thinking that the output of the device is one’s own thought. I call this the *Illusion of Directed Production* (IDP). I model the phenomenality of contentful DBS experiences alongside *auditory verbal hallucinations* (AVHs) and *thought insertions* (TIs) to show the potential for delusion formation and metacognitive and self-narrating disturbances [2–4] when using such systems. I argue that psychological risks and worries are present in such systems regardless of whether they truly enhance cognition or extend the mind.

Keywords Brain-computer interface · Large language model · Deep brain stimulation · Mind-writing · Auditory verbal hallucination · Thought insertion

Introduction

Allen Coin and Veljko Dubljević discuss the possibility of using a *brain-computer interface* (BCI) to connect one’s brain to a program that can perform arithmetical calculations given user input as a kind of mental prosthetic for those who possess dyscalculia [1]. They define a conceptual model of such a device:

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Assumption 1. In order to facilitate instant, accurate, reliable, and constant transfer of data to and from the subject's brain in a portable form factor that would allow for everyday use, the design would likely need to be an invasive one with direct contact with the subject's cochlear nerve, visual cortex, or other perceptual regions of the brain, in order to transmit data to the subject through an implanted microelectrode array. Thus it would convert intention-driven neuronal activity into a control signal per designs allowing for the automatic control of external prosthesis...¹

Assumption 2. The device would be capable of reliably detecting and anticipating the subject's intention to perform a simple arithmetical calculation and could then translate this intention into a control signal to pass to the computational array.

Assumption 3. The device would read the control signal and perform the intended calculation utilizing a simple chip capable of calculating outputs of arithmetic equations like the ones found in simple calculators....²

Assumption 4. The device would transmit the result of the calculation to the subject in such a way that the subject could immediately mentally note, verbalize, or record the result³ [1].

C&D develop this model to help determine whether such technologies genuinely enhance cognition. They claim that it does. If this device would extend the mind in the way Extended Mind Theory posits it does [2], then it would produce genuine cognitive enhancement. C&D also argue that this device would have a positive impact on self-identity. They claim that such a device restores the users with dyscalculia to their "true self", similar to how anti-depressants are seen to restore the individual to their "true self" [3]. They likely take this to be the case because dyscalculia, like any cognitive deficiency, plausibly affects self-esteem and self-image negatively.

The model C&D develop is ethically interesting in ways they have not fully considered. What they describe is effectively a closed-loop deep brain stimulation

(DBS) system, the output signal of which is determined by input received by the brain via microelectrode arrays and modified algorithmically. What is of particular ethical interest is the nature of *contentful* DBS, or neural stimulations that produce meaningful sensations.

This essay conceives of a similar device that sends and receives signals more contentful than what C&D's model minimally requires. Suppose now that, instead of a calculator, the BCI connects to a Large Language Model (or any other advanced automated text producing software) and sends and receives natural language inputs and outputs. BCI technology already exists that allows for "brain-to-text communication", that is, devices that read brain waves and convert them to readable natural language outputs [4, 5]. Mind-writing technology is also emerging, as Beauchamp et al. have been able to produce visual impressions of letters and digits through sequential firing of microelectrodes on an array [6]. C&D offer no concrete technological analog for their fourth assumption, but Beauchamp et al.'s system would fulfill this role just fine. The following conceptual model for a **BCI-LLM system** defines four key processes for its operation analogous to the assumptions of C&D's model:

Process 1: The BCI anticipates when the brain-to-text communication apparatus is intended to be used. In Card et al.'s system, the device anticipates this based on the attempt to produce speech [5].

Process 2: The BCI converts brainwave signals to natural language in the way Card et al [5] have achieved in their brain-to-text communication device.

Process 3: The natural language output is sent as a prompt to some automated text-producing algorithm. This could be an LLM such as ChatGPT, or a more sophisticated algorithm.

Process 4: The transformed output is sent back to the BCI. It is then converted into a signal delivered through DBS which provokes a meaningful thought-like experience ("mind writing").

It should be noted that the kind of DBS required for sending and receiving signals would be different and involve different parts of the cortex. Nevertheless, a system with both features would allow for seamless natural language communication between brain and computer. It is highly plausible that in the future, such devices would emerge, become more advanced, readily available, and adapted for a variety of consumer and commercial applications. For example, those looking to enhance

¹ C&D envision their device to operate in the way BCI technology developed by Hochberg & Colleagues [41] does.

² C&D cite a patent for a device that functions like a calculator that the BCI will connect to [42].

³ While C&D do not cite a technology that would fit this role, a recently developed DBS device and stimulation paradigm could fulfill this role, described in Sect. 1.

their thinking in a way that automates certain mundane thought processes and research tasks more efficiently than with using external interfaces leaves them more time to engage in more complex information-synthesizing operations. Such a system would plausibly meet this demand. While LLMs are the most sophisticated form of automated text production at the time of writing, this essay refers to “BCI-LLM system” as any system where the brain directly interacts with any sort of automated text-producing software.

This essay will assess the potential psychological risks of using BCI-LLM systems for the purpose of cognitive “enhancement”.⁴ There are two chief sources of concern: the input detection system, which could read “false positives”, and the contentful DBS hardware and stimulation paradigm (protocols and methods of hardware use). These are both independently potential sources of *metacognitive disturbance*. When taken together, the *apparent output spontaneity* of the former and the potential *phenomenal indistinctness from thought* of the latter can combine to produce disturbances akin to *auditory verbal hallucinations* (AVHs) or *thought insertions* (TIs). AVHs and TIs are hallmark experiences of psychotic conditions like schizophrenia. Both are experiences that are perceived as “external” and not identified as coming from the self. BCI-LLM systems can plausibly produce outputs that challenge one’s sense of self and sense of control in a similar way that AVHs and TIs do. This essay argues that it is plausible that related psychotic symptoms or even conditions may emerge from the use of such a device, regardless of whether true cognitive enhancement or mind extension occurs.

One type of metacognitive disturbance BCI-LLM systems can produce is what I call the *Illusion of Directed Production* (IDP). IDP’s are perceptions that lead to the misattribution of the cause of cognitive processes and their correlated physical outputs to oneself. Once one realizes IDP is occurring or has occurred, trust in one’s own metacognition and self-reporting abilities would likely

decrease, leading to dysregulation of the ability to reliably report one’s own inner state (“metacognition”). Likewise, anomalous experiences had by persons with psychotic or pre-psychotic conditions produce similar challenges.

This essay provides the theoretical basis for understanding IDP and the conditions for its manifestation as well as other metacognitive disturbances BCI-LLM systems could produce. Section “[BCI-AI and BCI-LLM Systems](#)” defines the BCI-LLM system and the technologies that could instantiate it. Section “[BCI-LLMs and The Illusion of Directed Production \(IDP\)](#)” describes IDP and the conditions for its manifestation. Section “[The Psychological Risks of BCI-LLM Systems](#)” analyzes the possible psychological risks using BCI-LLM systems by first modelling the phenomenality of its outputs, then using Carruthers’s model to assess its capacity for metacognitive disturbance, and then finally assessing its capacity for producing psychosis related symptoms or conditions. I conclude with a discussion on its implications for theoretical psychology, AI ethics, and potential directions for further research.

BCI-AI and BCI-LLM Systems

A “BCI-AI System” is a theoretically possible closed-loop deep brain stimulation (DBS) system where the DBS output signal is dependent upon inputs received directly from the brain via microelectrode arrays. These inputs are transformed in some way by deep learning AI. The DBS system is “closed-loop” because stimulation is dependent upon certain conditions (the presence of an input), rather than open-loop systems where DBS is given independently of any brain state. It is, broadly construed, a system which combines both brain-to-text communication technology (sometimes referred to as “mind-reading” technology) and “mind-writing”⁵ technology.

⁴ It is debatable whether the use of such a device *actually* enhances cognition, or in the terms of Extended Mind Theory [2], the pairing of brain and BCI-LLM system is merely “cognitive distribution”, where the two parts operate too independently from each other to be considered a true “coupled system”. Sven Nyholm [15] argues in the same vein that, regardless of whether the device is inside or outside the body, the subject only becomes “artificially intelligent” when using it. It is outside the scope of this essay to argue for one position or the other in this debate.

⁵ “Mind-reading” should not be interpreted according to the popular impression that mind-reading necessarily implies having one’s most inner secrets exposed, but rather should be understood as simply technology that allows the formation of information circuits directly from the brain to some digital interface. Nevertheless, it is plausible ethical problems of “mental integrity” are present even in this narrow sense. Likewise with “mind-writing” technology. It should not be understood as technology that directly changes one’s beliefs or creates new memories, but is simply technology which, through electrical stimulation in the brain, produces some meaningful neural effect. For a detailed study of the “hype” and therefore misunderstanding such terms generate, see Gilbert & Russo 2024 [43].

A BCI-AI System could target a variety of brain systems for the purposes of enhancing their functionality, whether it is visual or aural imagination, mathematical reasoning, etc. This essay focuses on the production of text or speech – both to narrow and simplify the discussion to a single medium and because natural language is a paradigmatic example of informationally rich, contentful outputs which are perhaps most illustrative of the ethical worries of this system. A BCI-LLM system is a kind of BCI-AI System which would serve as an aid to the enterprises of the inner monologue through the automated production of text. The output is perhaps experienced as speech, perhaps as thought. This system would in theory bestow greater freedom to the agent to perform more complex cognitive operations than was possible without automation, such as philosophy and advanced theoretic reasoning rather than more “menial” tasks such as data retrieval than was possible without automation. It does so in a way that some theorists like C&D believe would “extend” the mind or “enhance” cognition [1]. Insofar as AI is “intelligent”, this “intelligence” would seem to be shared with or becomes a part of the user.

There are reports that DBS systems can affect the personality and augment or transform the sense of self of some patients in unintended ways [7–9]. Some studies indicate patients who had undergone DBS therapy experience post-operative self-estrangement or other personality changes, both positive and negative, as observed in the treatment of Parkinson’s Disease [8, 10], indicating that their sense of self was expanded or otherwise positively transformed during treatment due to an increase of perceived control over their life.

Additionally, “intelligent BCI systems” are also reported to cause changes in the personality and sense of self [11, 12]. Frederic Gilbert et al. describe a clinical trial of six epileptic patients who had an “intelligent BCI system” implanted. This system would monitor the brain and alarm the patient using an external device if the neural preconditions for a seizure were detected. Upon detection, appropriate proactive measures can be taken to prevent a seizure. Four patients reported no change in their sense of self. However, one patient reported an expanded or enhanced sense of self, while another reported the exact opposite: a sense of self-estrangement and loss of control *because* of the intelligent BCI system. When the devices were explanted, the patient that reported an expanded or enhanced sense of self experienced negative effects on their sense of self

the authors call “agential discontinuity” [12], similar to some patients whose DBS treatment had ceased.

This evidence supports the idea that such systems can have an unanticipated effect on the personality and/or the self, for better or worse. These effects occur *even without* contentful DBS. As will be shown below, contentful DBS poses its own unique psychological risks.

The BCI-LLM System

If the output of C&D’s device is delivered in such a way that the “...subject could immediately mentally note, verbalize, or record the result”, then how exactly will the agent experience the output? Contentful DBS protocols are emerging that can meet the demand of the kind of system D&B model. For example, Beauchamp et al.’s “dynamic current steering method” [6] is used in a DBS system that produces visual impressions of letters through sequential stimulation of microelectrodes on an array.

This system and protocol could produce sufficiently contentful outputs for C&D’s system – in this case, digits perceived visually. In the case of the BCI-LLM system, however, the demand for much richer content requires a more sophisticated *stimulation paradigm* and hardware. A stimulation paradigm is *how* hardware is used to produce some effect. In Beauchamp et al.’s contentful DBS system, their discovery was that the sequential firing of microelectrodes on the visual cortex led to artificial visual impressions. Evolving DBS protocols alongside hardware improvements would allow for increasingly rich contentful DBS on different parts of the cortex. Depending on which cortical region is stimulated, linguistic contentful DBS could feel more thought-like or could be perceived visually or aurally as writing or speech, respectively.

A BCI-LLM system can be broken down into four distinct processes, analogous to the “assumptions” in C&D’s conceptual model. Each process will be individually discussed, and some preliminary ethical implications of each will be addressed.

Process 1

The BCI anticipates when the brain-to-text communication apparatus is intended to be used such as in Card et al. [5]. The BCI-LLM system must be able to determine *when* to read brain signals and send them to the LLM.

Card et al.'s neural speech prosthesis is one of the most successful at the time of writing that utilizes such a process. The system anticipates when the ALS patient with the prosthesis intends to speak. Card et al.'s system reads neural signals in the cortical region when the user attempts to speak. The system "...sustained 97.5% accuracy over a period of 8.4 months after surgical implantation, and the participant used it to communicate in self-paced conversations at a rate of approximately 32 words per minute for more than 248 cumulative hours" [5].

The team reports a false positive rate of less than 1%, meaning there are at least some cases where their system would detect input where none was intended. Therefore, outputs, if produced by unintended inputs, would be experienced as "spontaneous" insofar as the user cannot determine the correlating input. This could happen in such systems for a couple reasons. For example, system use would plausibly become second-nature by those who practice using it consistently over long periods of time. One could become so adept that its use becomes automatic or habitual: inputs are sent to it without the user recognizing an input was sent. The output would thus be unexpected. Secondly, it is possible that future devices may be more deeply integrated into the brain, accepting a greater depth of information, perhaps even unconscious cognitive processes the user is unaware of. It is thus conceivable that BCI outputs can possess apparent output spontaneity with unclear input origin. The false positive rate of input detection is a major ethical concern for systems which accept inputs in this manner, as will be discussed in later sections.

Process 2

The BCI converts brainwave signals to natural language in the way Card et al. [5] have achieved in their brain-to-text communication device. The brain-to-text communication technology developed by Card et. al is highly promising in its ability to allow ALS patients to communicate through reading only their brain signals when they were not able to before. There is also great potential for this technology to be used as a mechanism for text inputs to other BCI-compatible systems that receive text inputs. Card et al.'s system uses a chain of probabilistic algorithms that determine the most likely words intended to be spoken. While this is not a concern for this essay, bias in these algorithms could lead to misrepresented speech.

Process 3

The natural language output is sent as a prompt to some automated text-producing algorithm. This could be an LLM such as ChatGPT, or a more sophisticated algorithm. Once the text input is sent to an LLM, the input is transformed into content to be sent back through DBS or some other external means. The ethical issues of this process are identical to the ones of using LLM's in general such as algorithmic bias in LLM's [13, 14] and other epistemic considerations not relevant to this essay.

Process 4

The transformed output is sent back to the BCI. It is then converted into a signal delivered through deep brain stimulation ("mind writing" or text-to-brain communication). With sufficiently sophisticated DBS hardware such as microelectrode arrays and sufficient stimulation paradigms that can produce rich, contentful outputs, an output can be given to the subject in such a way that would "enhance" or "extend" native cognition. There is debate as to whether such a system would genuinely enhance or extend the mind in an appropriate way [1, 15]. C&D argue that it does, but Sven Nyholm disagrees. He believes the user of an LLM does not possess sufficient control of the output of the system for it to be considered to be their own.

Regardless of whether mind extension truly occurs, Process 1 and Process 4 possess significant ethical worry which will be the focus of the rest of the essay.

BCI-LLMs and The Illusion of Directed Production (IDP)

The ethical worry of contentful DBS experiences can be illustrated by a series of thought experiments. First, consider John Searle's "Chinese Room":

"Imagine a native English speaker who knows no Chinese locked in a room full of boxes of Chinese symbols (a data base) together with a book of instructions for manipulating the symbols (the program). Imagine that people outside the room send in other Chinese symbols which, unknown to the person in the room, are ques-

tions in Chinese (the input). And imagine that by following the instructions in the program the man in the room is able to pass out Chinese symbols which are correct answers to the questions (the output). *The program enables the person in the room to pass the Turing Test⁶ for understanding Chinese but he does not understand a word of Chinese*” [16].

There are two relevant factors in Chinese Room: (1) the *actual* locus and distribution of intelligence in the system (intelligence insofar as it is exhibited by processes functionally equivalent to internal cognitive ones), and (2) the *apparent* locus of intelligence to observers. In this scenario, the agent receiving input and output is merely following the instructions of the instruction book (analogous to software). The agent is totally alienated from the “intelligence” of the software and merely executes it. The instruction book is therefore the *actual* locus of intelligence. Likewise, the agent inside identifies the apparent locus of intelligence as the instruction book. However, the observers outside the room have no reason to believe that the output would come from anything other than another human. Given that the output is intelligently constructed, the locus of intelligence is placed upon an agent.

Nyholm considers his own version of the Chinese Room. In this version, the English speaker in the room is using an LLM that accepts Chinese characters as an input and produces appropriate outputs such that those outside of the room are also fooled into thinking that a fluent Chinese speaker is in the room [15]. The outcome for the outside observers is the same as in the original Chinese Room – they have no reason to believe that the output would come from anything other than a fluent Chinese speaker. One slight difference is that, unlike the instruction book of the original case, ChatGPT is a “black box” with a procedure completely hidden from the agent. The agent inside is no longer manually executing the program and is therefore even more alienated from the “intelligence” of the system. Nyholm concludes that the English speaker is, like in the original thought experiment, merely “artificially intelligent”, rather than cognitively enhanced.⁷ In other words, the fact that the English speaker is “artificially intelligent” is because the apparent locus of intelligence of the

observers outside the room differs from the actual locus of intelligence like in the original Chinese Room.

Now consider this version of the Chinese Room. In this version, there is no room, nor any external equipment, only an individual with an implanted BCI-LLM system hidden from sight. The cognitive system is just as hidden from the actual fluent Chinese speakers as it is from the person who interacts with this system. Like the first two cases, the intelligence, or at least the cognitive processes involved, is contained in the black-box LLM. However, another question emerges. *Could the agent operating the system be fooled, too?*

If the inputs and outputs are in Chinese, then the agent can reflect on their own lack of any knowledge of Chinese and conclude that the algorithm is certainly the one producing the answers. However, if the output was exclusively in a language they were fluent in, the agent could misinterpret both that the output is their own thoughts *and* that the output reflected their own epistemic state. In other words, there may be instances in this version of the Chinese Room when the apparent and actual locus of intelligence is mismatched *for the agent*. This may seem counterintuitive at first since the “black box” nature of the LLM affords a sense of alienation to its algorithm and hence one would be disinclined to identify the output of the algorithm as their own. But normal cognition and creative processes are also “black box” systems that agents with brains identify as their own. Creative processes are often spontaneous and not strictly controlled. Furthermore, the perceived “internality” of the BCI-LLM system affords the belief that what process is happening is their own and therefore under their control. I call this the Illusion of Directed Production (IDP henceforth). IDP is caused by perceptions that lead to the misattribution of the cause of cognitive processes and their correlated outputs (text, art, etc.) to oneself. There are two phenomenal variables which are most relevant to IDP in BCI-LLM systems: *phenomenal distinctness* of the output from natural thought and *apparent output spontaneity* (Fig. 1).

The details of how such a BCI-LLM system would operate and how its outputs would be experienced depend on the input detection system, the implanted DBS hardware, and the stimulation paradigm used. The ability to have strict control of the input should not be

⁶ The “Turing Test” evaluates whether a machine can think like a human based on human observation of the output [44].

⁷ There are a growing number of cases reported where the locus of intelligence even at this level is misplaced by the agents using LLMs.

presumed, as input detection systems could sometimes pick out brain signals not intended to be inputs. There are other reasons outputs can be perceived as spontaneous. For example, device use could become second-nature to the user, leading to input signals the user is unaware of. Or, the device could exhibit “deep integration” and erroneously interpret cognitive processes that the user is unaware of as input. These cases could be especially jarring since the input would be related to private thoughts which they may not even notice.

It is unclear which stimulation paradigm will be the best or most feasible for long-form natural language outputs. Nevertheless, it is plausible that contentful DBS could be thought-like or speech-like. If the BCI produces phenomenally distinct outputs, then the agent can tell apart their own thoughts from the output just by “feeling” them apart. Any accompanying sensations of the BCI-LLM output, for example, would be indications that the thought-experience is unnatural. This “feeling” differs from other kinds of contentful DBS characteristics, such as semantic structure, writing style, and the type of content itself, that may differ from one’s natural thought. Such differences could certainly manifest between

contentful DBS and one’s own thoughts, and such differences can help determine whether an internal experience results from one’s natural cognition or is the result of DBS.

The relationship between the BCI-LLM user and the apparent output spontaneity of the system is also important. As mentioned above, it could be the case that inputs can be sent to the device without awareness of it occurring. There is certainly reason to be worried about this sort of technology if it is the case that both (1) the outputs are phenomenally indistinct from one’s own thoughts, and (2) some outputs are apparently spontaneous. The two together would lead to experiences where one receives outputs that resemble regular thoughts without being aware that an input was sent.

This would be the most favorable condition for generating IDP. The only possible evidence an agent could use to distinguish such an output from their own thoughts is the form and content of the output itself. If the contentful DBS was in Chinese, for example, and the agent knew no Chinese, then the agent could easily determine that the Chinese

Fig. 1 Table mapping phenomenal distinctness of output and spontaneity of input to the potential risk of IDP

		Low Output Spontaneity	High Output Spontaneity
Phenomenally Indistinct Output	Phenomenally Indistinct Output	• Easily controlled inputs • While output resembles thoughts, they are easily linked to intention to use the LLM • Low risk of IDP	• Inputs can be entered unintentionally • Output resembles other thoughts • High risk of IDP
		• Easily controlled input • Output appears “other” • Very low risk of IDP	• Inputs can be entered unintentionally • Output appears “other” • Low risk of IDP

thoughts were not their own, even if they were apparently spontaneously generated and the output “feels” the same as their own thoughts. However, if the LLM produces outputs in their own language and with content concerning a topic that they are somewhat but not completely familiar with, then the false attribution of certain outputs to their own cognitive and epistemic states (such as beliefs and knowledge) could occur.⁸ If this is done and the agent later realizes that what they thought about their epistemic state is false, then a metacognitive disturbance occurs. Metacognitive disturbances need not manifest only through IDP, as the next section will show. The consequences of experiencing these disturbances could lead to adverse psychological effects.

The Psychological Risks of BCI-LLM Systems

Coin & Dubljević characterize the BCI system they envision to “restore” self-identity to its users in the same way that antidepressants are thought to [3]. For them, integration and use of this device is a *therapy* that (at least apparently) eliminates a deficiency in the user’s cognition. What will become clear by the end of this section is that a more advanced version of this system, such as a BCI-LLM system, has the potential to *disturb* one’s sense of self. This is certainly not apparent nor likely if only the limited scope of potential outputs (limited to numerical outputs) and presumed low input spontaneity of C&D’s proposed device are considered. For example, people with dyscalculia who would use C&D’s device would likely not encounter IDP since they lack the capacity to naturally form mathematical thoughts or execute mathematical calculations. Therefore, it is safe to assume users of C&D’s device would not confuse the output with their own thoughts.

BCI-LLM systems have the potential to cause metacognitive disturbances and self-disruptions akin to how *Auditory verbal hallucinations* (AVH) and *thought insertions* (TI) have been theorized to [17, 18]. AVHs and TIs are anomalous experiences

commonly had by people with psychotic conditions, particularly schizophrenia. The apparently spontaneous contentful DBS experience would also be an anomalous experience, but it is one whose output is, at face value, “external” in the sense that AVHs and TIs are.

This section uses theories and models of AVHs and TIs to take a closer look at what experiencing contentful DBS produced by a BCI-LLM system is like and what the possible consequences of the chronic use of it is. Section 3.1 models the phenomenality and potential delusionality of BCI-LLM outputs by adopting and revising an integrative phenomenological model of AVHs and TIs developed by Clara Humpston & Matthew Broome [19]. Section 3.2 describes how Glenn Carruthers’s metacognitive model of the sense of agency [17] and how this sense is disturbed does well to explain how contentful DBS experiences might disturb this sense and lead to metacognitive dysregulation. Section 3.3 describes how the subjective cognitive disturbances produced by BCI-LLM systems could lead to the development of pre-psychotic syndromes (the psychotic prodrome⁹) or psychotic conditions according to the “phase model”¹⁰ of the development of schizophrenia. This section concludes that it is possible that chronic exposure to contentful DBS can cause acute psychosis related symptoms that subside from disuse, or at worst, cause a chronic psychotic condition.

Modelling the Phenomenality of Contentful DBS Experiences

AVHs and TIs are similar insofar as they are anomalous experiences and hallmark symptoms of individuals with psychotic conditions. But they do differ phenomenally. Humpston & Broome (H&B) develop an “integrative model” of TIs and AVHs which map them and other kinds of thoughts and experiences using two variables: *audibility* (using a spectrum ranging from ‘silent’ to ‘audible’) and *agency/ownership* (using a spectrum ranging from ‘internal’ to

⁸ It is also plausible that, through routine exposure to contentful DBS, some thoughts will conform to the structure of the content more frequently, making contentful DBS more difficult to distinguish from natural thought.

⁹ The psychotic prodrome is an early-onset syndrome of psychotic conditions typically diagnosed only in retrospect due to their subclinical nature and difficulty of early detection [34, 35].

¹⁰ [31–33].

‘external’) [19]. Figure 2 is a first attempt at mapping the range of thought experiences produced by BCI-LLMs onto H&B’s model.

Experiences which exhibit agency include “own thought”, or the natural inner monologue, and “inner speech/verbal thought”, which captures the ability to reproduce voices using the imagination. The “external” side of the agency spectrum are where H&B place TIs, ‘soundless voices’, and ‘true’ AVHs, all of which are anomalous experiences that appear outside of the agent’s control with varying degrees of audibility. Contentful DBS experiences could be more speech-like or more thought-like depending on the stimulation paradigm and hardware. Therefore, the range of possible BCI-LLM experiences extends across the entire audibility spectrum.

At first glance, the range of contentful DBS’s produced by BCI-LLM systems would begin at the “external” side of the agency/ownership spectrum and end at a position that is not fully internal. After

all, the LLM operates independently from the mind. Even so, its output is not entirely independent since it operates based on the user’s prompts, even in cases where the output appears spontaneous. There is always some influence that the internal mental state has on the state of the BCI-LLM system.

A couple issues immediately arise when mapping possible BCI-LLM thought-experiences onto H&B’s model in this way. Firstly, there is no position on the model to properly map IDP. IDP is a result of confusing a thought that comes from an *actual* external source to be from an “internal” or natural source. IDP is an illusion that affords the production of a false judgment about the nature of the thought-experience in question. It is only in retrospect that someone, after metacognitive deliberation, would identify the error, if at all.

This issue is related to the second: BCI-LLM outputs are not necessarily of a delusional character in the way that the model presumes TIs and AVHs to

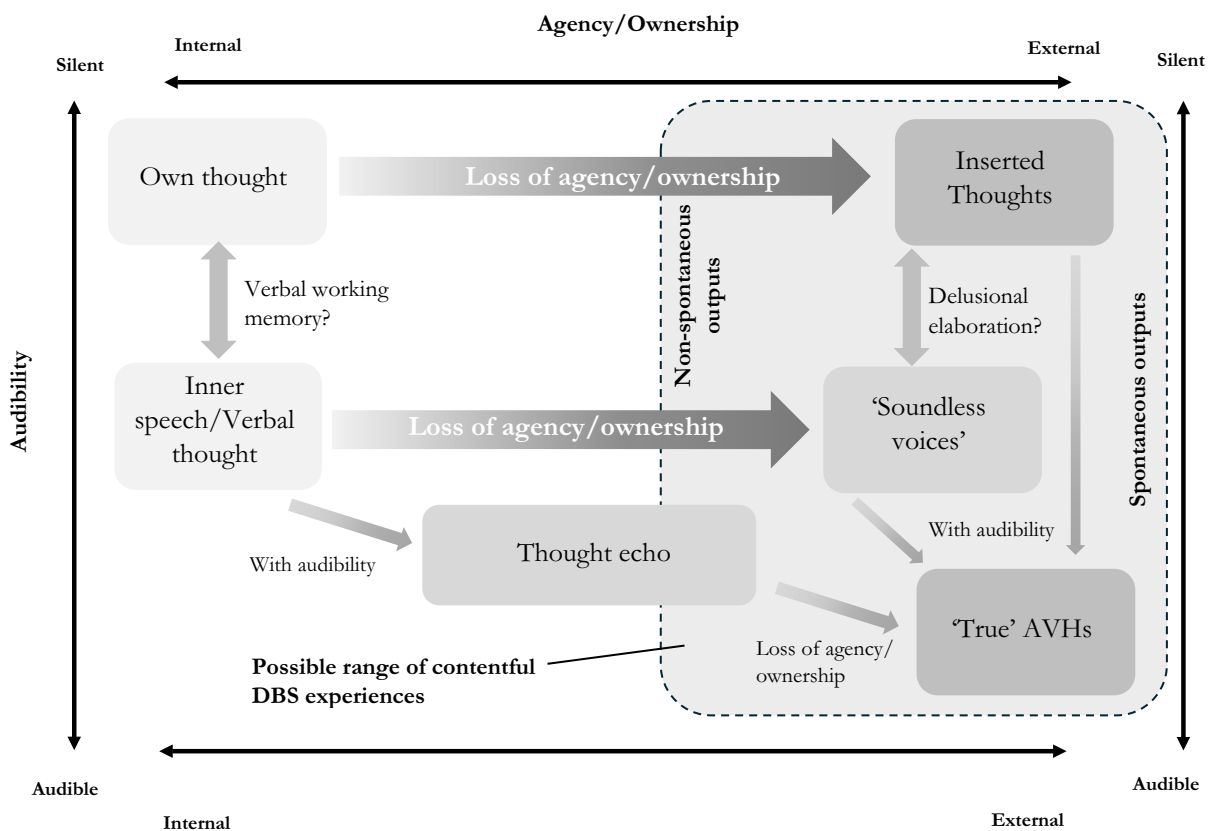


Fig. 2 A first attempt at mapping the range of linguistic contentful DBS experiences using Humpston & Broome’s integrative model of AVHs and TIs [19]

be. Yet, they are still “external”. H&B’s model presumes delusionality of thought experiences perceived as “external”. This is a safe assumption to make when natural thought is solely relevant, as all natural thought-experiences have an internal origin. Therefore, interpreting a thought-experience as “external” in this context normally indicates some failure to appreciate its internal origin. However, with experiences of *true external origin* in the picture, this assumption is no longer safe to make.

To accommodate H&B’s model to fit both IDP and all possible contentful DBS experiences, the “internal” and “external” ends of agency/ownership must be clarified to “apparent internality” and “apparent externality”, respectively, with “agency/ownership” renamed to “*apparent* agency/ownership” or “*apparent* origin” (“apparent” insofar as the experience affords judgment or is perceived in some way to be of either “internal” or “external” origin). Secondly, a third variable must be added, “*actual* agency/ownership” or “*actual* origin”, which ranges from “actual internal” to “actual external” (Fig. 3). In this way, experiences of BCI-LLM outputs can be modelled alongside TIs and AVHs without confusion. Figure 3 illustrates how a mismatch in apparent and actual origin represent false judgments of one’s mental state, indicating a failure in metacognition, as will be discussed further in 3.2.

Figure 2 presupposes “external” experiences to necessarily imply delusionality. However, BCI-LLM outputs are neither necessarily delusional nor are the experiences themselves false beliefs necessary for the presence of delusion. Another question then emerges: how does delusionality fit into the experiences of TIs, AVHs, and IDP? Doxastic accounts of TIs understand delusion as an epistemic state. Under doxastic accounts of delusion, a delusion is a false belief with some degree of pathological motivation for adherence that typically lacks appropriate justification [20]. There are two main doxastic accounts of TIs. First is the *endorsement model*, which claims that “...abnormal experience comprises the very content of delusional beliefs” [21]. Advocates of the endorsement model argue “...that the relevant abnormal experience comprises the content of the delusional beliefs reported, so patients simply believe or endorse what they have experienced” [21–24]. That is, there is some vague content within the experience of the

thought, taken as a belief, that gives it its “inserted” character. On the other hand, the *explanationist model* of TIs argues “...that subjects adopt delusional beliefs as a way of explaining abnormal experiences that do not have the same content of the underlying doxastic state” [21, 25]. Explanationists contend that delusions are a *consequence* of bad judgments about anomalous or unusual experiences.

The explanationist model is a stronger candidate for explaining IDP and the experience of apparently spontaneous BCI outputs. This is primarily because BCI outputs are not themselves delusions, nor are they beliefs of the user. This is not to discredit the endorsement model, since Pablo López-Silva suggests that it is possible both models work for different kinds of TIs [21]. It is important to note that the judgment of internality/externality under an explanationist model comes *after* the anomalous experience. The false belief necessary for a delusion to form can be modelled with Fig. 3 as a conflict between *apparent* and *actual* thought ownership and origin. IDP is a case where the thought experience in question comes from an external source that appears internal. A TI under the explanationist model is apparently of external origin but was *actually* of internal origin.

Metacognitive Disturbances of BCI-LLM Systems

Conflict between apparent and actual origin of a thought-experience signals that a *metacognitive disturbance* has occurred. This is when one’s account of one’s own epistemic state does not capture reality and thus leads to false belief production and possibly also to delusion.

“Metacognition” has been characterized as our ability to “think about thinking” [26]. It has also been suggested that metacognition is a faculty which aids in the production of narrative selfhood [18]. While selfhood has been characterized as minimally an ipseity or some essential feature that makes one who they are, the “narrative self” account makes no substantive claim about the self or selfhood [27]. The narrative self is a “social self” which has “...characteristics such as social identity, personality, habits, style, personal history, etc.” [27]. These sorts of narratives are essential for providing a foundation for making decisions in the world. Indeed, such narratives also give a picture of what one’s beliefs and

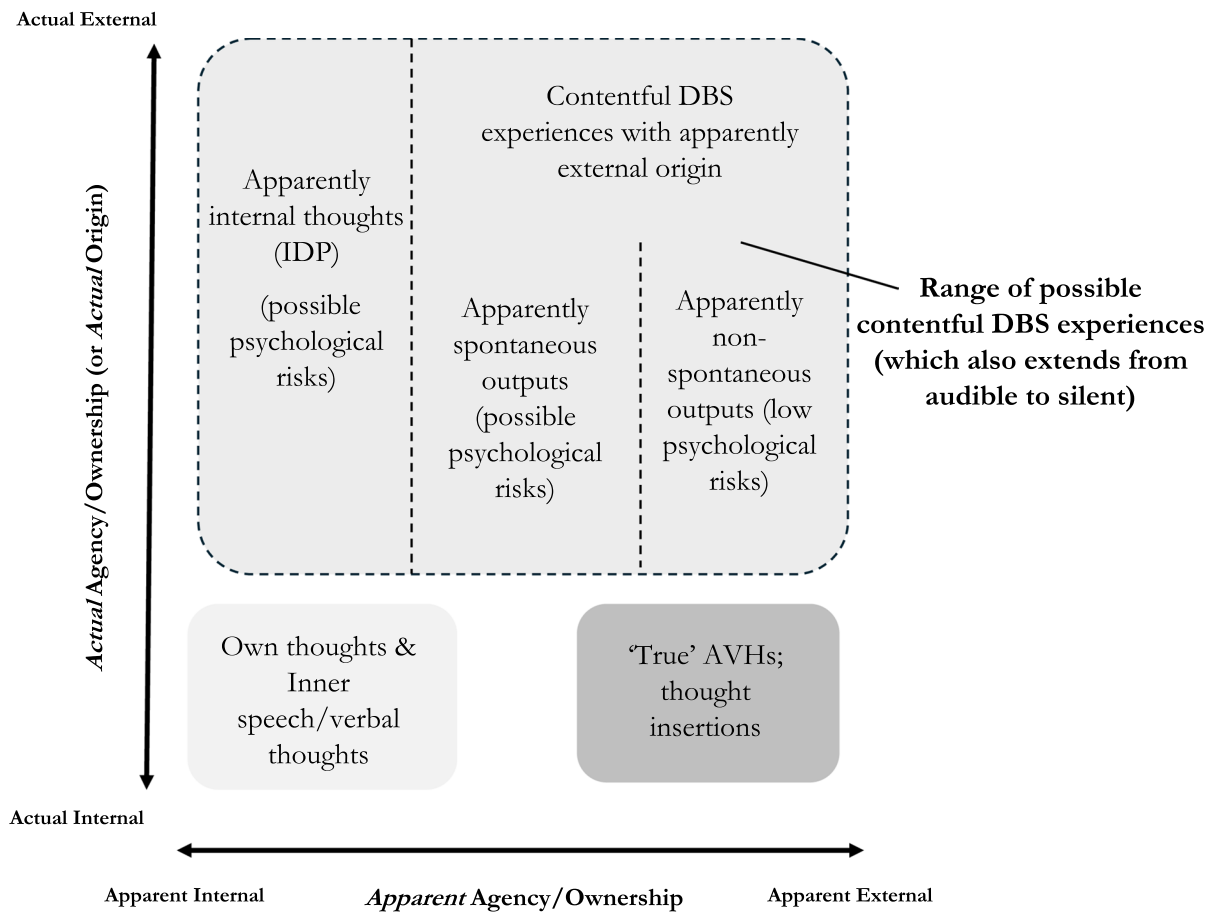


Fig. 3 Various thought-experiences mapped with respect to *actual* agency/ownership and *apparent* agency/ownership

other epistemic states are – metacognition is required for this. A metacognitive disturbance is an event that thus complicates the production of self-narratives.

The possibility of a BCI-LLM system’s ability to produce metacognitive disturbances and how it can produce negative reactions in the user can be illustrated by the following scenario, *Disturbance I*:

1. A contentful DBS is mistaken as one’s own thought (IDP). The BCI-LLM system produced a thought-like experience through an input the user did not intend.
2. The output is taken to be representative of one’s beliefs and knowledge, or is otherwise taken to be one’s own thought process which unintentionally leads to new belief formations. The agent makes decisions in the world based on this new belief.

3. The agent comes to realize that they possess contradictory beliefs or at least conflicting epistemic states. One of them must be rejected.
4. The agent concludes that the BCI has produced one of these states which they did not realize were not their own.
5. Distrust in one’s ability to produce coherent metacognitive narratives of their own beliefs develops as a result of repetitions of 1–4.
6. A loss of a sense of agency is sustained. Neurotic reactions such as anxiety and depression may result as a consequence.

This type of disturbance causes the individual to doubt their capacity to produce accurate metacognitions of their own thoughts and thus produce a fully sound picture of themselves. Notice that in this case, the BCI-LLM system can produce any sort of output for

this disturbance to occur, so long as the topic is somewhat related to beliefs and epistemic states of an individual. Distress plausibly results when the agent realizes they fail to give good accounts of their epistemic states as they used to. Metacognitive disturbances are not limited to IDP, as false judgments similar to those necessary for TIs can manifest in other ways:

Disturbance II: Someone receives an apparently spontaneous BCI output like *Disturbance I*, but due to its unusual or triggering nature it is interpreted to be caused by *someone else's interference*. This person could explain that their device was likely hacked, or some other external influence (God or demons for example) is at play.

Disturbance III: Someone's own thoughts are taken to be an apparently spontaneous BCI-LLM output when no output is produced. This could occur due to unusual or triggering content produced by natural thought, or because the thought in question, by the BCI-LLM's influence, has some features which make it resemble a BCI-LLM output.

Both *Disturbance II* and *Disturbance III* plausibly erode one's own sense of agency in thought production and disturbs metacognition to a similar degree as the first. *Disturbance II*, unlike *Disturbance I*, occurs independently of contentful DBS. A system with an external output could produce this disturbance, too, since the explanation serves to "deflect" or deny the triggering content. *Disturbance III* could occur in chronic BCI-LLM system users, especially if, through routine exposure to its outputs, their natural thoughts start to bear similarities in structure to contentful DBSs.

Someone who already suffers from a psychotic condition may be more predisposed to experience these kinds of disturbances. However, it is plausible that one need not have any such predisposition for these disturbances to occur. Glenn Carruthers developed a model which explains how metacognitive disturbances, particularly those like TIs and AVHs, disturb a sense of agency over thoughts [17].

In Carruthers's model of properly working metacognition, *monitoring representations* "...represent specific object-level processes as they occur" [17]. They are representations of normal thinking. Metacognitive beliefs are "...beliefs or theories

about how object-level processes work" [17]. On his account, metacognitive beliefs regulate object-level processes. He cites findings that women who believe women are worse at math than men are worse at math than women who believe otherwise [28]. That is, there may be a causal connection between metacognition about one's skill (whether or not it is based on a stereotype) and the performance of that skill. The sense of agency is a product of the whole process working properly, yielding self-narratives that are satisfactory enough for one to use to describe oneself and use as a basis for decision making.

Carruthers asserts that, if metacognition is treated as an object-level process itself, then "...a deficit in monitoring representations of these processes will lead to deregulation of thought generation" [17]. That is, a disturbance in one's sense of agency over their thoughts dysregulates thought generation *in general*. Carruthers focuses on our ability to inhibit or suppress thought production as a key ability which produces a sense of agency over thoughts. When someone begins to develop unusual metacognitive beliefs, or they begin rejecting their previous metacognitive beliefs in the case of retroactively recognizing IDP, then monitoring representations become more inadequate at capturing what is actually occurring at the object level. This leads to further change of metacognitive beliefs. This could continue repeating in a positive feedback loop that further exacerbates the disturbance of one's sense of agency.

Carruthers's account can be slightly changed to accommodate the unique phenomena of IDP and contentful DBS by instead asserting that metacognitive disturbance arises from an *insufficiency* of metacognition, rather than a *deficiency*. The average person should not be expected to produce sound metacognitive beliefs about atypical experiences – this is true of TIs, AVHs, and contentful DBS. The standard for producing sound metacognitive beliefs and self-narratives that include contentful DBS experiences would likely be *higher* than what is typically needed when concerned with only natural thought. Methods for developing sufficient metacognitive resilience for dealing with contentful DBS experiences would therefore be a valid approach for allowing safer use of BCI-LLM systems. Such accounts are outside the scope of this essay, but the feasibility of related

methods such as metacognitive therapy for psychotic conditions have been demonstrated [29, 30].

BCI-LLM, Psychosis, and the Psychotic Prodrome

There are worrying implications if BCI-LLM systems would indeed cause metacognitive disturbances as the models above indicate. Some models of the etiology of psychotic disorders suggest anomalous experiences are what trigger their development [31–33]. Peter Handest et al. produce a synthesis of the “phase models” of both Klaus Conrad and Joachim Klosterkötter that traces the development and onset of psychosis to initial subjective cognitive disturbances [33]. In this model, disturbances of memory, concentration, and other subjective cognitive disturbances (often subclinical in their first manifestations) lead to increased “hyperreflexivity”. This term refers to “...heightened awareness of aspects of experience that are normally tacit or implicit” [27]. That is, one becomes far more attentive of their inner state in order to better preserve the soundness of their metacognitions and self-narratives. This in turn causes dramatic shifts in thought-patterns which then exacerbate the issue further. These new thought patterns are described as “thought interference, pressured thinking, thought-block, and obsessive-like preservation”. In order to maintain the soundness of self-narratives, the agent is thought to make great effort to preserve the apparent interiority and control of their thoughts.

This in turn leads to further increase in hyperreflexivity which leads to what these authors call “transparency”, characterized by Handest et al. to be a dissociation and disidentification with one’s own thoughts as the subjective/internal world and objective/external world become increasingly difficult to distinguish. This model finally suggests that the culmination of this process results in a relinquishment of self-identification with the subjective world, or the world of thought, and as a result it is as displaced from the self as the external world. The result is the experience of “internal” phenomena as “other”, hence full blown AVHs and TIs become a common experience for these individuals. Schizophrenia thus fully manifests.

What is worrying about this model is that some contentful DBS experiences would likely be considered the kinds of “subjective cognitive disturbances”

that are theorized to be the initial cause of psychosis. Like AVHs, TIs, and their theorized etiological precursors, the mismatch between apparent and actual origin of thought experiences leads to metacognitive disturbances. The early stage of having these anomalous experiences is referred to as the *psychotic prodrome*, which is “...a period of pre-psychotic disturbance, representing a deviation from a person’s previous experience and behavior” [34]. The prodromal phase is constituted by subclinical signs that can last from weeks to years [35]. BCI-LLM systems thus possess this potential risk and worry. It is a worry that should inform development and testing of such systems and related devices.

Even if one does not accept this model, the dysregulation of metacognition potentially caused by BCI-LLM systems would nevertheless be a distressing experience that leads to a worsening quality of life. It is certainly possible that in healthy individuals, the negative effects of its use would subside with cessation of use. Some research of the psychopathology of such experiences shows how depression, for example, are associated with the self-disturbances which result as a consequence from the kinds of “subjective cognitive disturbances” relevant to the phase model [36, 37].

Conclusion

Implications for Theory, Ethics, and Research

Even though Coin & Dubljević envision their device to have only positive effects as a result of the cognitive “enhancement” it gives, the reality is that contentful DBS experiences, especially in the context of natural language outputs, could cause unanticipated effects on one’s sense of self, similar to non-contentful DBS technology. Indeed, such devices could plausibly cause metacognitive disturbances that lead to negative psychological, and plausibly also psychosis-related syndromes. Therefore, caution should be taken when developing and testing these systems, especially since no animal model can adequately assess the psychological safety of BCI-LLM systems, or any text-to-brain communication technology, generally speaking. One approach to mitigate risks in human testing is to run psychological background checks on individuals who have a history or whose family has a history

of psychotic conditions. Or, if such systems are developed for an unregulated consumer market outside of a medical setting, it may be better to only work with subjects who have already implanted such a device to themselves.

There are other implications for BCI-LLM systems not dealt with in this essay that further research can shed more light on, described below.

IDP and bias in AI

It is possible that an agent could lack the metacognitive ability to detect IDP. In this case, IDP would thus never be recognized. Or there are cases where IDP is recognized but is consistently suppressed by an individual, who continually rationalizes that it is indeed their own thought. In these cases, no metacognitive disturbances ever occur in the ways described in Sect. 3. The consequence of this is that the subject uncritically accepts the content of the outputs to be representative of their own epistemic states. This is worrisome because LLMs can exhibit biases in the outputs, such as through data selection bias in the compilation of training data [13, 14]. What this means is that subjects who uncritically accept outputs as epistemic states will also uncritically accept these biases as a feature of their own cognition, which may include epistemic states on political matters. This is already an issue with AI devices with external user interfaces, as some users will uncritically accept LLM output as fact, even though “truth” is not a criterion involved in the training of LLM systems. BCI-LLM systems would possess an even higher degree of authority for some users, especially those with narcissistic tendencies.

This issue is a microcosm of what Helga Nowotny calls the “illusion of control” when relying on AI [38]. For Nowotny, relying on AI to accomplish more and more tasks may give us the sense that we are obtaining greater power. She argues that this comes at the expense of our ability to directly control the consequences of its processes [38, 39]. Thus, the world and our lives will be impacted in ways we may not intend. If Nowotny’s claim that our automation-enamored egos occlude an uncertain future is correct, then similar worries should be had about automating cognitive processes in general, and this essay demonstrates a potential

manifestation of this illusion. Theorizing more about the political ramifications of automating cognitive processes seems to be a fruitful research direction.

BCI-LLM Systems, Metaepistemology, and Metacognitive Therapy

Concluded with the remark that developing sufficient metacognitive resilience as a method for allowing the safe use of BCI-LLM systems, perhaps adapted from a kind of metacognitive therapy used for patients with schizophrenia [30]. It is possible that some healthy individuals could naturally develop their own resilience to the metacognitive disturbances produced by BCI-LLM systems. That is, the metacognitive challenges presented by BCI-LLM systems could afford adaptations in metacognition. By paying closer attention to the content and experience of one’s thoughts (known as hyperreflexivity) one could develop either (1) sounder metaepistemological states organized by specific principles, or (2) deeper awareness of the minute phenomenal details of their thought-experiences. There is likely a rich ground for interdisciplinary research between conceptual ethics, metaepistemology, and normative epistemology on one hand (as defined by McPherson & Plunkett [40]), and the development of metacognitive therapies which seek to instill more robust metacognitive states on the other. The potential psychological risks of BCI-LLM system use gives further incentive for such research to be pursued.

Other Empirical Research Directions

With the possession of such a technology, psychologists and psychiatrists could be able to determine to what extent (a) AVHs and TIs serve as adequate models for BCI-LLM outputs, and if so, then (b) which models of the etiology of psychosis are best able to explain psychotic conditions. A BCI-LLM system could allow for more controlled experiments with external-feeling thoughts. Of course, such empirical research would be highly ethically dubious, given the risks that such a technology may entail. Assessing anecdotal reports of device use may be the best option to answer these research questions to get a

better picture not only of the nature of BCI-LLM systems, but of AVHs and TIs as well.

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